EE4-07 Coding Theory

Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

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1. (a) Let $\mathcal{C} \subset \mathbb{F}_3^7$ be a linear code. Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 2 & 0 & 2 & 1 & 2 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 & 2 & 0 & 2 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_3^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_3^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

iii What is the distance of the code defined by $\boldsymbol{H}$? Justify your answer. [4]

iv Assume that a message $\boldsymbol{m}_1$ is encoded into a codeword $\boldsymbol{c}_1$ using $\boldsymbol{G}$. The codeword $\boldsymbol{c}_1$ is transmitted over a ternary symmetric channel. Let the received word be

$$\boldsymbol{y}_1 = [1\ 1\ 1\ 0\ 0\ 2\ 1]\,.$$

Compute the syndrome vector $\boldsymbol{s}_1$. Find the output of the minimum (Hamming) distance decoding, say $\hat{\boldsymbol{c}}_1$. [2]

v Assume that a message $\boldsymbol{m}_2$ is encoded into a codeword $\boldsymbol{c}_2$ using $\boldsymbol{G}$ The codeword $\boldsymbol{c}_2$ is transmitted over a ternary erasure channel. Let the received word be

$$\boldsymbol{y}_1 = [1\ 0\ ?\ ?\ 1\ 0\ 2]\,.$$

Set the question marks in $\boldsymbol{y}_2$ to zero and compute the corresponding syndrome vector $\boldsymbol{s}_2$. Find the transmitted codeword $\boldsymbol{c}_2$. [2]

Solution:

i

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 2 & 2 & 1 \end{bmatrix}.$$

ii

$$\boldsymbol{G} = \begin{bmatrix} 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 2 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 1 & 0 \\ 2 & 2 & 2 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

iii The code distance is 3. It is easy to verify that every two columns of $\boldsymbol{H}$ are linearly independent. There exist three columns that are linearly dependent, for example, columns 1 2 and 4.

iv It can be verified that

$$\boldsymbol{s}_1 = [2\ 1\ 0]$$
$$\hat{\boldsymbol{c}}_1 = [1\ 1\ 1\ 1\ 0\ 2\ 1]$$

v It can be verified that

$$\boldsymbol{s}_2 = [1\ 2\ 1]$$
$$\boldsymbol{c}_2 = [1\ 0\ 2\ 2\ 1\ 0\ 2]$$

(b) Let $f(x) = x^3 + x + 1 \in \mathbb{F}_3[x]$.

i Let $g_1(x) = x^2 + 1 \in \mathbb{F}_3[x]$. Find the *monic* polynomial $h_1(x) = \gcd(f(x), g_1(x)) \in \mathbb{F}_3[x]$ using Euclidean algorithm. Based on Bezout's identify, find $a_1(x), b_1(x) \in \mathbb{F}_3[x]$ such that $h_1(x) = a_1(x)f(x) + b_1(x)g_1(x)$. [3]

ii Let $g_2(x) = x^2 + 2 \in \mathbb{F}_3[x]$. Find the *monic* polynomial $h_2(x) = \gcd(f(x), g_2(x)) \in \mathbb{F}_3[x]$ using Euclidean algorithm. Based on Bezout's identify, find $a_2(x), b_2(x) \in \mathbb{F}_3[x]$ such that $h_2(x) = a_2(x)f(x) + b_2(x)g_1(x)$. [3]

Solution: It can be verified that

$$\begin{aligned} h_1(x) &= \gcd(f(x), g_1(x)) = 1 \\ &= f(x) + 2x * g_1(x). \end{aligned}$$

and

$$h_2(x) = \gcd(f(x), g_2(x)) = x + 2$$
$$= 2 * f(x) + x * g_2(x).$$

(c) Consider the finite field $\mathbb{F}_{16}$ detailed below

	$\mathbb{F}_2[z]/z^4+z+1$		$\mathbb{F}_2[z]/z^4+z+1$
0	0	z^7	z^3+z+1
1	1	z^8	z^2+1
z	z	z^9	z^3+z
z^2	z^2	z^{10}	z^2+z+1
z^3	z^3	z^{11}	z^3+z^2+z
z^4	$z+1$	z^{12}	z^3+z^2+z+1
z^5	z^2+z	z^{13}	z^3+z^2+1
z^6	z^3+z^2	z^{14}	z^3+1

i Compute $\beta_1 = (z^3+z)\cdot(z^3+z^2+z) \in \mathbb{F}_{16}$. [2]

ii Compute $\beta_2 = (z^3+z^2+z)^{-1}\cdot(z^3+z^2) \in \mathbb{F}_{16}$. [2]

iii Compute the polynomial $f(x) = (x-\beta_1)\cdot(x-\beta_2) \in \mathbb{F}_{16}[x]$. Show key steps of computations. [3]

Solution: It can be verified that

$\beta_1 = z^9 \cdot z^{11} = z^5 = z^2 + z.$

$\beta_2 = z^{-11} \cdot z^6 = z^{-5} = z^{10} = z^2 + z + 1.$

$f(x) = (x - z^5)(x - z^{10}) = x^2 - (z^2 + z + z^2 + z + 1)x + 1 = x^2 + x + 1.$

2. Consider the prime number $p = 131$ and a primitive element $g = 54$ of $\mathbb{F}_p$.

(a) Find $g^n \bmod p$ where $n = 1,\ 2,\ 4,\ 8,\ 16,\ 32,\ 64,\ 128$, respectively. Write your final answer in a table format as below.

$n =$	1	2	4	8	16	32	64	128	
$g^n \equiv$	?	?	?	?	?	?	?	?	mod p

[4]

(b) Based on the results of the above calculations, find $x = \log_g(114) \bmod p$. [1]

Solution: It can be computed that

$n =$	1	2	4	8	16	32	64	128	
$g^n \equiv$	54	34	108	5	25	101	114	27	mod p

Hence

$$x = \log_g(114) = 64.$$

(c) Alice and Bob would would like to communicate using the ElGamal public key cryptography which involves the prime number p and the primitive element g. Suppose that the private keys that Alice and Bob choose are $k_{s,a} = 72$ and $k_{s,b} = 96$, respectively.

i Find the public keys that Alice and Bob will use respectively. [2]
Note: In the following sub-questions, Alice and Bob only know each other's public key (not the other person's private key).

ii Suppose that Bob needs to send Alice a plaintext message $m = 73$. He chooses a random number $t = 80$ to add randomness to his message. Find the encrypted message that Bob will use. Show key steps of your computations. [5]

iii Suppose that Alice sends Bob an encrypted message $(S_1, S_2) = (55, 50)$. Find the plaintext message from the encrypted message. Show key steps of your calculations. [5]

Solution: Alice's public key is

$$A = g^{72} = g^{64+8} \equiv 114 * 5 \equiv 46 \bmod p.$$

Bob's public key is

$$B = g^{96} = g^{64+32} \equiv 114 * 101 \equiv 117 \bmod p.$$

The encrypted message from Bob to Alice is given by

$$\begin{aligned}(g^t, mA^t) = (g^{80}, 73 * 46^{80}) &= (g^{64+16}, 73 * 46^{64+16}) \\ &\equiv (99, 73 * 112) \equiv (99, 54) \bmod p.\end{aligned}$$

Bob can decrypt the message from Alice by computing

$$S_1^{-k_{s,b}} * S_2 = 55^{-96} * 50 \equiv 55^{34} * 50 \equiv 29 \bmod p.$$

(d) In this sub-question, consider another prime number $p = 107$.
Let $a = 51$ and $b = 56$. Find their orders in $\mathbb{F}_p$, denoted by $\text{ord}(a)$ and $\text{ord}(b)$ respectively. Which numbers are primitive elements in $\mathbb{F}_p$? Justify your answer. [8]

Solution: The number $p-1 = 106$ can be factorized into $p-1 = 106 = 2 * 53$. For any element $\alpha \in \mathbb{F}_p$, its order $\text{ord}(\alpha)$ can only take values in $\{1, 2, 53, 106\}$. It can be verified that

$$\begin{array}{rccccl} n = & 1 & 2 & 53 & 106 & \\ a^n \equiv & 51 & 33 & 106 & 1 & \bmod p \\ b^n \equiv & 56 & 33 & 1 & 1 & \bmod p \end{array}$$

As a consequence, $\text{ord}(a) = 106$ and a is a primitive element. $\text{ord}(b) = 53$ and b is not a primitive element.

3. Consider codes on $\mathbb{F}_q$. The code size (the number of codewords) is denoted by M, the length of codewords is denoted by n, and distance of the codebook is denoted by d. If the code is linear, its dimension is denoted by k.

(a) i Let $q = 3$, $n = 10$, and $d = 5$.
What does Hamming bound predict about M and k?
What does Gilbert-Varshamov bound predict about M and k?
Show key steps of your computations. [4]

ii Let $q = 11$, $n = 10$, and $d = 5$.
What does Hamming bound predict about M and k?
What does Gilbert-Varshamov bound predict about M and k?
Show key steps of your computations. [4]

Solution: The volume of a ball on $\mathbb{F}_q$ is given by

$$V(r) = \sum_{i=0}^{r} \binom{n}{i} (q-1)^i.$$

Consider the case where $q = 3$. Hamming bound requires

$$V(2) = 201,\ M \leq q^n/V(2) \approx 293, k \leq 5.17$$

Gilbert-Varshamov bound says that there exists a code such that

$$V(4) = 4521,\ M \geq q^n/V(4) \approx 14, k \geq 2.40$$

Consider the case where $q = 11$. Hamming bound requires

$$V(2) = 4601,\ M \leq q^n/V(2) \approx 5.64e6, k \leq 6.48$$

Gilbert-Varshamov bound says that there exists a code such that

$$V(4) = 2.22e6,\ M \geq q^n/V(4) \approx 1.17e5, k \geq 3.91$$

(b) Consider field $\mathbb{F}_p$ with $p = 7$ and a primitive element $g = 3$.

i Let $n = 6$ and $d = 5$. What does Singleton bound predict about k? [1]

ii The above Singleton bound can be attained by a Reed-Solomon code $\mathcal{C}$.

It can be defined by its parity check matrix of the form

$$\boldsymbol{H} = \begin{bmatrix} 1 & g & \cdots & g^{n-1} \\ 1 & g^2 & \cdots & g^{2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & g^{n-k} & \cdots & g^{(n-k)(n-1)} \end{bmatrix}.$$

Compute the parity check matrix $\boldsymbol{H}$ of the Reed-Solomon code that attains what you computed above. This will allow you to do decoding below. [4]

iii Suppose that a codeword $\boldsymbol{x} \in \mathcal{C}$ is transmitted and the received word is given by

$$\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{e} = [4\ 6\ 3\ 2\ 3\ 1].$$

Compute its syndrome vector $\boldsymbol{s}$. Find the corresponding syndrome polynomial $S(z)$. [5]

iv Using the key equation

$$E(z) = L(z) * S(z) \bmod z^{d-1},$$

find the corresponding error locator polynomial $L(z)$ and error evaluator polynomial $E(z)$. [5]

v Using the locator polynomial $L(z)$ find the error locations. [2]

Solution: It is clear

$$k = n + 1 - d = 2.$$

The parity check matrix is given by

$$\boldsymbol{H} = \begin{bmatrix} 1 & 3 & 2 & 6 & 4 & 5 \\ 1 & 2 & 4 & 1 & 2 & 4 \\ 1 & 6 & 1 & 6 & 1 & 6 \\ 1 & 4 & 2 & 1 & 4 & 2 \end{bmatrix}.$$

The syndrome vector is given by

$$\boldsymbol{s} = \boldsymbol{y} * \boldsymbol{H}^{\mathsf{T}} = [1\ 5\ 1\ 1];$$

Hence the syndrome polynomial is given by

$$S(z) = x^3 + x^2 + 5x + 1.$$

Apply Euclidean algorithm to the key equation.

$$\begin{aligned} z^4 &= q_1(z)S(z) + r_1(z) \\ S(z) &= q_2(z)r_1(z) + r_2(z) \end{aligned}$$

where

$$\begin{aligned} q_1(z) &= x + 6 \\ r_1(z) &= 3x^2 + 4x + 1 \\ q_2(z) &= 5x + 3 \\ r_2(z) &= 2x + 5 \end{aligned}$$

Note that

$$\begin{aligned} E'(z) = r_2(z) &= S(z) - q_2(z)(z^4 - q_1(z)S(z)) \\ &= (1 + q_1(z)q_2(z))S(z) - q_2(z)z^4 \\ &= L'(z)S(z) - q_2(z)z^4. \end{aligned}$$

We have

$$\begin{aligned} L'(z) &= 5x^2 + 5x + 5 \\ E'(z) &= 2x + 5. \end{aligned}$$

This results in

$$\begin{aligned} L(z) &= x^2 + x + 1 \\ E(z) &= 6x + 1. \end{aligned}$$

It can be verified that

$$L(g^{-2}) = L(4) = 0 \text{ and}$$
$$L(g^{-4}) = L(2) = 0.$$

Hence the error locations are 2 and 4 (locations start from 0).